

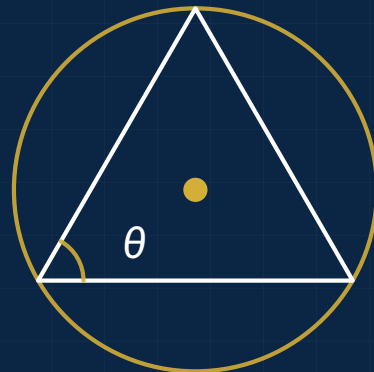


DR ESLAM AHMED
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CHAPTER 4 • GEOMETRY & MEASURES • TOPIC 3

Bearings, Scale & Constructions

A bearing is measured from North, clockwise, in three figures. Scales turn cm into real distances. Constructions are arcs you leave on the page.



5

VARIANTS

12

PAST-PAPER PARTS

29

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Bearings & back bearings

V2 Scale drawings & maps

V3 Perpendicular bisector

V4 Angle bisector

V5 Bearings with trig

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Bearings & constructions at a glance



Your notes

Bearings, Scale & Constructions

DIRECTION & DISTANCE

V1 Bearings & back bearings

$$149 + 180 = 329$$

PAST PAPERS: 1 OF 12

V2 Scale drawings & maps

$$1 \text{ cm} : 200 \text{ m}$$

PAST PAPERS: 2 OF 12

RULER & COMPASSES

V3 Perpendicular bisector

arcs from A and B

PAST PAPERS: 2 OF 12

V4 Angle bisector

arcs from B, then the arms

PAST PAPERS: 4 OF 12

WITH TRIGONOMETRY

V5 Bearings with trig

cosine rule, sine rule

PAST PAPERS: 3 OF 12

THE KEY HABIT

NORTH LINE AT EVERY POINT

Draw North where you stand, then turn clockwise.

★ How often does each variant come up?

11 past-paper questions (Jan 2020 → Nov 2025) hold 12 parts; 29 practice parts — grouped by type.

	PAST PAPERS (OUT OF 12)		PRACTICE PARTS
V1 Bearings, back bearings	● ● ● ● ● ● ● ● ● ● ● ●	1	9
V2 Scale drawings, maps	● ● ● ● ● ● ● ● ● ● ● ●	2	9
V3 Perpendicular bisector	● ● ● ● ● ● ● ● ● ● ● ●	2	5
V4 Angle bisector	● ● ● ● ● ● ● ● ● ● ● ●	4	0
V5 Bearings with trig	● ● ● ● ● ● ● ● ● ● ● ●	3	6

WHAT THIS TELLS YOU A 2-mark construction (usually an angle bisector) is often Question 2–6 — free marks if the arcs are left on. Bearings with the sine/cosine rule are the 5–6 mark questions later in the paper.

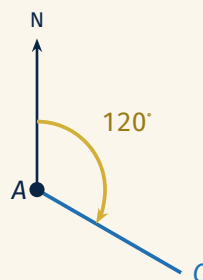
BEFORE YOU START

The big idea

**Three rules for every bearing.**

From North, clockwise, three figures
(045°, not 45°).

“The bearing of C **from** A”: stand at A, face North, turn clockwise until you face C.


Bearing facts
Back bearing
 $\pm 180^\circ$

Under 180: add.
Over 180: subtract.

Due east
 090°

South 180° , west
 270° .

North lines

parallel

Co-interior
angles add to
 180° .

Three figures
 008°

Always 3 digits.

Constructions: what they give you
Perpendicular bisector of AB

equal distance from A and B

Points the same distance from two points.

Angle bisector of ABC

equal distance from both arms

Points the same distance from two lines.

**CALCULATOR FIRST, THEN CHECK**

Leave every arc on the page — no arcs, no marks, even if the line is perfect.

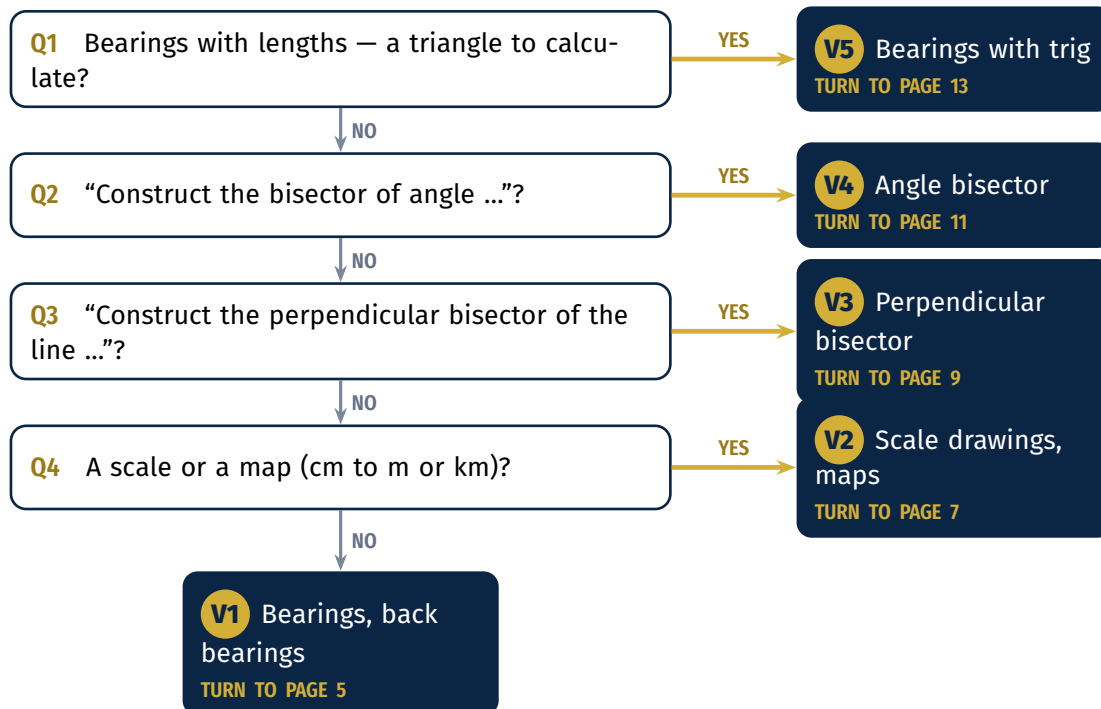
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Bearings & Back Bearings

Back bearing: add 180° (or subtract if over 180°).

SEEN IN PAST PAPERS

1 / 12

PRACTICE QUESTIONS: 9

 Your notes

Came up in: May 2021 P2H Q3

? What is it?

In plain words

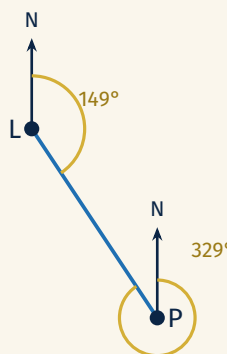
The way back is the **opposite direction**: turn round by **180°** .

Bearing under 180° : **add** 180. Over 180° : **subtract** 180. The North lines at the two points are parallel, so this always works.

IT LOOKS LIKE

149° → 329°

255° → 075°



How do I do it?

1 Read

Bearing of what, from where?

2 Under 180?

Add 180.

3 Over 180?

Subtract 180.

4 Three figures

Write 075°, not 75°.

$\pm 180^\circ$ North lines are parallel.

V1 Bearings & Back Bearings

VARIANT 1 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"The bearing of Paris from London is 149° . Bearing of London from Paris?" → $149 + 180$

"bearing of Q from P is 255° " → $255 - 180 = 075^\circ$

Two bearings linked by b → **Back bearing** = $b \pm 180$: **equation**

WORKED EXAMPLE **PAST PAPER**

May 2021 · Paper 2H · Q3 · 2 marks

The bearing of Paris from London is 149° . Work out the bearing of London from Paris.

STEP 1 $149 < 180$ ◀ under 180

STEP 2 $149 + 180 = 329$ ◀ add 180

ANSWER 329° ✓ $329 - 149 = 180$ ✓

WATCH OUT

✗ $180 - 149 = 031^\circ$

✓ $149 + 180 = 329^\circ$

The back bearing is 180° more, not the difference from 180.

EXAM TIP

Sketch the two North lines — the co-interior angles make it obvious.

Check yourself

calculator first · answers on page 15

1.1

Dr Eslam

The bearing of B from A is 064° . Find the bearing of A from B.

Answer:

1.2

Dr Eslam

The bearing of Q from P is 255° . Find the bearing of P from Q.

Answer:

1.3

Practice · Hard

The bearing of B from A is b° ; the bearing of A from B is $6b^\circ$. Find the bearing of A from B.

Answer:

V2

VARIANT 2

VARIANT 2

Scale Drawings & Maps

Real = map $\times$ scale; map = real $\div$ scale; 1 : n needs the same units.

SEEN IN PAST PAPERS

2 / 12

PRACTICE QUESTIONS: 9



Your notes

Came up in: Jan 2022 P2H Q5

What is it?

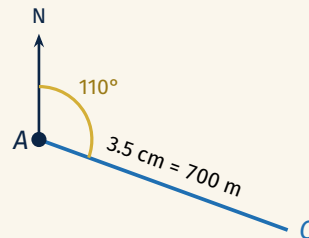
In plain words

“1 cm represents 200 m”: every cm on the map is 200 m in real life. A 700 m path is $700 \div 200 = 3.5$ cm.

Scale 1 : n : change to **the same unit**
 – 200 m = 20 000 cm, so 1 : 20 000.

IT LOOKS LIKE

1 cm : 200 m

 $1 : n$ 

How do I do it?

- | | | |
|---|------------------------|-------------------------------------|
| 1 | Scale | What does 1 cm represent? |
| 2 | Map $\rightarrow$ real | $\times$ the scale. |
| 3 | Real $\rightarrow$ map | $\div$ the scale. |
| 4 | 1 : n | Both in cm: n = real cm for 1 cm. |

SAME UNITS FOR 1: n 1 m = 100 cm, 1 km = 100 000 cm.

V2 Scale Drawings & Maps

VARIANT 2 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Mark the position of C ... 700 m on a bearing of 110° "

→ $700 \div 200 = 3.5$ **cm on 110°**

"Write the scale of the map in the form $1 : n$ "

→ **Same units:** $1 : 20\,000$

"Measure ... how far in real life?"

→ **Measure in cm, $\times$ scale**

WORKED EXAMPLE **PAST PAPER**

January 2022 · Paper 2H · Q5 · 4 marks

Scale: 1 cm represents 200 m. House C is 700 m from A on a bearing of 110° . (a) How do you mark C? (b) Write the scale in the form $1 : n$.

STEP 1 $700 \div 200 = 3.5$ cm ◀ map length

STEP 2 Measure 110° clockwise from North at A; mark C 3.5 cm along ◀ (a) protractor and ruler

STEP 3 $200 \text{ m} = 20\,000 \text{ cm}$ ◀ same units

STEP 4 $1 : 20\,000$ ◀ (b)

ANSWER (a) 3.5 cm on 110° (b) $1 : 20\,000$ ✓ $3.5 \times 20\,000 = 70\,000 \text{ cm} = 700 \text{ m}$ ✓

WATCH OUT

✗ $1 : 200$

✓ $1 : 20\,000$

1 cm and 200 m are different units — change m to cm.

EXAM TIP

Use a sharp pencil and measure the bearing from the North line at the right point.

Check yourself

calculator first · answers on page 15

2.1

Dr Eslam

1 cm represents 250 m. A path is 3.4 cm on the map. Real length? Scale as $1 : n$?

Answer:

2.2

Dr Eslam

Map scale $1 : 50\,000$. A road is 6 cm on the map. Real length in km?

Answer:

2.3

Dr Eslam

Map scale $1 : 60\,000$. Two towns are 4.2 km apart. Distance on the map?

Answer:



VARIANT 3

Perpendicular Bisector

Same compass width from A and from B
(more than half AB); join the two crossings.

SEEN IN PAST PAPERS

2 / 12

PRACTICE QUESTIONS: 5

2 / 12

 Your note

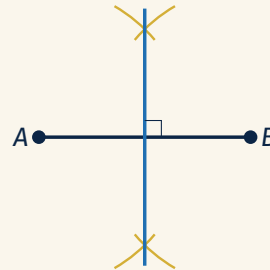
Came up in: Jan 2020 P2HR Q6 • Jun 2025 P2H Q4

What is it?

In plain words

The perpendicular bisector cuts AB **in half** at 90° . Every point on it is the **same distance** from A and from B .

Compasses wider than half of AB ;
arcs above and below from A , then
the **same width** from B .



IT LOOKS LIKE

construct the perpendicular bisector of AB

How do I do it?

1 Width

Compasses more than half of AB .

2 From A

Arc above and below the line.

3 From B

Same width: arcs cross the first ones.

4 Join

Rule through both crossings. Keep the arcs.

SAME WIDTH, KEEP THE ARCS 2 marks: arcs + line.

V3 Perpendicular Bisector

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO*"Construct the perpendicular bisector of the line AB"*→ **Arcs from A and B, same width***"Show all your construction lines"*→ **Never rub the arcs out****WORKED EXAMPLE****PAST PAPER**

June 2025 · Paper 2H · Q4 · 2 marks

Using ruler and compasses only, construct the perpendicular bisector of the line AB. Show all your construction lines.

STEP 1 Open the compasses to more than half of AB ◀ width

STEP 2 From A, then (same width) from B: arcs ◀ arcs above and below that cross

STEP 3 Rule a line through the two crossing points ◀ join

ANSWER a line through the crossings, at 90° to AB through its midpoint

✓ arcs left on ✓

**WATCH OUT**

✗ changing the width between A and B

✓ the same width from both

Different widths give a line that is not at the midpoint.

**EXAM TIP**

Check with a ruler: your line should cut AB exactly in half.

**Check yourself**

calculator first · answers on page 15

3.1

Dr Eslam

What is true of every point on the perpendicular bisector of AB?

Answer:

3.2

Dr Eslam

AB = 8 cm. Where does its perpendicular bisector cross AB, and at what angle?

Answer:

3.3

Dr Eslam

AB = 6 cm. How wide must the compasses be?

Answer:

V4**VARIANT 4 MOST COMMON IN THE EXAM**

Angle Bisector

Arc from the vertex across both arms; from each crossing, equal arcs that meet; join to the vertex.

SEEN IN PAST PAPERS

**4 / 12**

PRACTICE QUESTIONS: 0

Your notes

Came up in: Jan 2020 P2H Q5 • Oct/Nov 2020 P1HR Q2 • Nov 2024 P1H Q2 • Nov 2025 P1H Q3

? What is it?

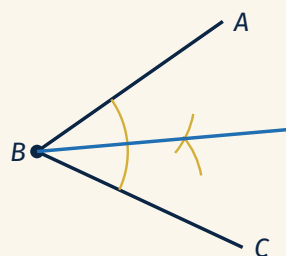
In plain words

The angle bisector cuts the angle **exactly in half**. Every point on it is the **same distance** from both arms.

One arc from the **vertex**, then two equal arcs from where it crossed the arms.

IT LOOKS LIKE

construct the bisector of angle ABC



How do I do it?

1 From B

One arc cutting both arms.

2 From each crossing

Same width: two arcs that meet inside the angle.

3 Join

Rule from B through the meeting point.

4 Keep

All arcs stay on the page.

VERTEX, ARMS, JOIN The most common construction.

V4 Angle Bisector

VARIANT 4 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Construct the bisector of angle ABC ” → **Arc from B , then from both arms**

“Using ruler and compasses only” → **No protractor**

“points the same distance from BA and BC ” → **The angle bisector**

WORKED EXAMPLE PAST PAPER

November 2024 • Paper 1H • Q2 • 2 marks

Use ruler and compasses only to construct the bisector of angle ABC . You must show all your construction lines.

STEP 1 Point at B : one arc that crosses BA and BC ◀ from the vertex

STEP 2 From where it crosses BA : an arc inside the angle ◀ arm 1

STEP 3 Same width, from where it crosses BC : an arc crossing it ◀ arm 2

STEP 4 Rule from B through the crossing point ◀ join

ANSWER a line from B splitting angle ABC into two equal angles

✓ arcs left on ✓

WATCH OUT

✗ measuring with a protractor

✓ arcs only

“Ruler and compasses only” — a protractor scores 0.

EXAM TIP

Make the first arc large — small arcs give a wobbly line.

Check yourself

calculator first • answers on page 15

4.1

Dr Eslam

What is true of every point on the bisector of angle ABC ?

Answer:

4.2

Dr Eslam

Angle $ABC = 70^\circ$. What angle does the bisector make with BA ?

Answer:

4.3

Dr Eslam

Angle $PQR = 128^\circ$. Find the angle between its bisector and QR .

Answer:

V5

VARIANT 5

Bearings with Trigonometry

Draw North at each point, find an angle inside the triangle, then cosine / sine rule.

SEEN IN PAST PAPERS

●●●●●●●●●● 3 / 12

PRACTICE QUESTIONS: 6

3 / 12



Your notes

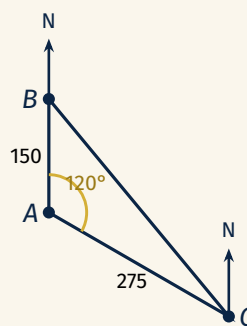
Came up in: Jan 2021 P1H Q16 · Nov 2023 P1H Q18 · Nov 2024 P2H Q24

What is it?

In plain words

Put the bearings on a sketch with a **North line at each point**. Use **co-interior** angles (add to 180°) and angles at a point to find an angle **inside** the triangle.

Then the **cosine rule** for a side, the **sine rule** for another angle, and turn it back into a bearing.



IT LOOKS LIKE

$$AB = 150$$
$$AC = 275$$
bearing 120°

How do I do it?

1 Sketch

North line at every point.

2 Inside angle

From bearings: co-interior, ± 180 , at a point.

3 Rules

Cosine rule $\rightarrow$ side; sine rule $\rightarrow$ angle.

4 Back to a bearing

Clockwise from North at the right point.

NORTH AT EVERY POINT Then it is a triangle question.

V5 Bearings with Trigonometry

VARIANT 5 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO*"Calculate the bearing of C from B"*→ **Angle at B inside the triangle, then from North***Two sides and a bearing between them* → **Cosine rule for the third side****WORKED EXAMPLE****PAST PAPER**

January 2021 · Paper 1H · Q16 · 5 marks

B is due north of A, $AB = 150$ m. C is 275 m from A on a bearing of 120° . Calculate the bearing of C from B.

STEP 1+2

$$BAC = 120^\circ; BC^2 = 150^2 + 275^2 - 2(150)(275)\cos 120^\circ, BC = 373.3$$

◀ inside angle; cosine rule

STEP 3

$$\frac{\sin B}{275} = \frac{\sin 120^\circ}{373.3} \Rightarrow B = 39.6^\circ$$

◀ sine rule

STEP 4

$$180 - 39.6 = 140.4$$

◀ from North at B (facing south is 180°)**ANSWER** 140°

✓ coordinates give the same ✓

**WATCH OUT**✗ bearing = 39.6° ✓ $180 - 39.6 = 140^\circ$

39.6° is measured from south (along BA), not from North.

**EXAM TIP**

Keep full calculator values until the final bearing.

**Check yourself**

calculator first · answers on page 15

5.1

Nov 2024 P2H

$AB = 8.4$ m on a bearing of 067° from A; $BC = 9.2$ m on a bearing of 129° from B. Bearing of A from C?

Answer:

5.2

Nov 2023 P1H

$AB = 3.6$ km on 054° , then $BC = 8.4$ km on 132° , then back to A, at 6 km/h. Total time?

Answer:

5.3

Dr Eslam

B is 6 km east of A; C is 8 km due south of B. Bearing of C from A?

Answer:

CHECK YOURSELF

Answers



Every answer below was checked in Python (check_answers.py).

1.1 244° **1.2** 075° **1.3** 216° **2.1** 850 m; 1 : 25 000**2.2** 3 km**2.3** 7 cm**3.1** equidistant from A and B**3.2** 4 cm from A, at 90° **3.3** more than 3 cm**4.1** equidistant from BA and BC**4.2** 35° **4.3** 64° **5.1** 280° **5.2** 3 h 38 min**5.3** 143.1°

THE WHOLE TOPIC ON ONE PAGE

Bearings cheat sheet



Your notes
V1 Back bearing 1/12

$$149 + 180 = 329$$

 ± 180 ; three figures.
V2 Scale 2/12

$$1 \text{ cm} : 200 \text{ m} = 1 : 20\,000$$

Same units for $1 : n$.
V3 $\perp$ bisector 2/12

same width from A, B

Equidistant from two points.

V4 Angle bisector 4/12

vertex, arms, join

Equidistant from two lines.

V5 With trig 3/12

cosine, then sine rule

North at every point.

**TOP 3 LOST MARKS**

1. arcs rubbed out ×
2. bearing measured anticlockwise ×
3. $1 : 200$ with mixed units ×

BOTTOM LINE North, clockwise, three figures · ± 180 back · same units for $1 : n$ · keep the arcs.